

What are the risks of artificial intelligence (AI)?

[Source: <https://www.tableau.com/data-insights/ai/risks>]

AI is a specific branch of computer science concerned with mimicking human thinking and decision-making processes. These programs can often revise their own algorithms by analyzing data sets and improving their own performance without needing the help of a human. These are often programmed to complete tasks that are too complex for non-AI machines.

There are some risks associated with AI, some pragmatic and some ethical. Many of these are purely hypothetical situations that may happen in the future without proper precautions, and some are real concerns that we deal with today.

Real-life risks include things like consumer privacy, legal issues, AI bias, and more. And the hypothetical future issues include things like AI programmed for harm, or AI developing destructive behaviors.

Real-life AI risks

1. Privacy
2. AI bias
 - a. Data bias
 - b. Societal bias
3. Human Interactivity
4. Legal responsibility

1. Privacy

One of the biggest concerns experts cite is around consumer data privacy, security, and AI. Many companies had already data privacy violations with their collection and use practices, and experts worry this may increase as we start utilizing more AI. Another major concern is that there are currently few regulations on AI (in general, or around data privacy) on the national or international level. The EU introduced the “AI Act” in April 2021 to regulate AI systems considered of risk; however, the act has not yet passed.

2. AI bias

It’s a common myth that since AI is a computer system, it is inherently unbiased. However, this is blatantly untrue. AI is only as unbiased as the data and people training the programs. So if the data is flawed, impartial, or biased in any way, the resulting AI will be biased as well. The two main types of bias in AI are “data bias” and “societal bias.”

a. Data bias

Data bias is when the data used to develop and train an AI is incomplete, skewed, or invalid. This can be because the data is incorrect, excludes certain groups, or was collected in bad faith.

b. Societal bias

On the other hand, societal bias is when the assumptions and biases present in everyday

society make their way into AI through blind spots and expectations that the programmers held when creating the AI.

3. Human Interactivity

How does interacting with AI affect humans? There are physical safety concerns. In 2018, a self-driving car used by the rideshare company Uber hit and killed a pedestrian in a driving accident. If companies rely too much on AI predictions for when maintenance will be done without other checks, it could lead to machinery malfunctions that injure workers. Models used in healthcare could cause misdiagnoses.

And there are further, non-physical ways AI can harm humans if not carefully regulated. AI could cause issues with digital safety (causing defamation or libel), financial safety (this could be misuse of AI in financial recommendations, credit checks, or the opposite, such as complex schemes that steals or exploits financial information), or equity (biases built into AI that can cause unfair rejections or acceptances in a multitude of programs).

4. Legal responsibility

When something goes wrong, who is responsible? The AI itself? The programmer who developed it? The company that implemented it? Or, if there was a human involved, is it the human operator's fault? Different uses of AI will have different legal liabilities if something goes wrong.

Hypothetical AI risks

1. AI programmed for harm
2. AI develops destructive behaviors

1. AI programmed for harm

Another risk that experts cite when talking about the risks of AI is the possibility that something that uses AI will be programmed to do something devastating. The best example of this is the idea of "autonomous weapons" which can be programmed to kill humans in war.

2. AI develops destructive behaviors

Another concern, somewhat related to the last, is that AI will be given a beneficial goal, but will develop destructive behaviors as it attempts to accomplish that goal. An example of this could be an AI system tasked with something beneficial, such as helping to rebuild an endangered marine creature's ecosystem. But in doing so, it may decide that other parts of the ecosystem are unimportant and destroy their habitats. And it could also view human intervention to fix or prevent this as a threat to its goal.

Making sure that AI is fully and completely aligned to human goals is surprisingly difficult and takes careful programming. AI with ambiguous and ambitious goals are worrisome, as we don't know what path it might decide to take to its given goal.